



EAST WEST UNIVERSITY

Department of Mathematical and Physical Sciences (MPS)
Semester: Fall - 2025

Course Outline: MAT 205

Course Title : Linear Algebra and Complex Variables
Course Code : MAT 205
Pre-Requisite : MAT 102 and MAT 104
Credit Hour : 3
Course Instructor : Professor Dr. Md. Shariful Alam
Office Room : FUB 1004
Mobile Number : 01316814344
E-mail Address : shariful.alam@ewubd.edu
Class Schedule and Office Hour:

Day	Course No.	Section	Class Time & Room No.	Office Hour
S T	MAT 205	7	8:30AM-10:00AM, FUB-802	10.10 AM - 11.40 AM Room: FUB 1004
		8	11:50AM-1:20PM, FUB-604	

Course Goal:

For the students of any branch of science, knowledge of Mathematics is essential. Linear Algebra develops the basic concept on matrix, determinant, vector space, linear transformation and some other important topics, which will help the students to develop their analytical ability. Theories of Complex Variables are very important for all branches of Science and Engineering. This will help in developing the theories of Laplace Transformation, Fourier series and Fourier Transformation.

Learning Outcomes:

On successful completion of this course students will be able to

- solve systems of linear equations by using Gaussian elimination and Gauss-Jordan elimination method;
- be able to apply the basic techniques of matrix algebra, including finding the inverse of an invertible matrix using adjoint and row canonical form;
- understand the basic ideas of vector algebra: linear dependence and independence and spanning;
- be able to find the eigenvalues and eigenvectors of a square matrix using the characteristic polynomial and will know how to diagonalize a matrix when this is possible;
- understand the significance of continuity and differentiability for complex functions and be familiar with the Cauchy-Riemann equations;
- state and prove versions of Cauchy's theorem and its consequences including Cauchy's integral formula, the power series representation for analytic functions and the Fundamental Theorem of Algebra.

Course Contents : 1. Linear Algebra.
2. Complex Variables.

Description : The following topics will be covered throughout the semester.

Linear Algebra: Matrices and operations with matrices, Determinants, Systems of linear equations; Vector spaces; Linear independence; Basis and dimension; Linear transformations; Eigenvalues and eigenvectors; Inner Product Space, Diagonalization of matrices; LU-decomposition; Applications.

Complex Variables: Functions of a complex variable and their derivatives; Analytic function; Singularities; Cauchy integral theorem and formula; Power series and Laurent expansions; Calculus of residues and contour integrals; Applications.

Text book :

1. Elementary Linear Algebra-Applications Version, Howard Anton, Chris Rorres, Eleventh Edition, Wiley-India.
2. Complex Variables, Murray R Spiegel, Seymour Lipschutz, Schaum's Outline Series, Second Edition, McGraw-Hill.

Reference books :

1. Linear Algebra, 5th Edition, Seymour Lipschutz and Marc Lipson, Schaum's Series, McGraw-Hill.
2. Complex Variable by M.L. Khanna.

Term Examinations :

Mid Exam	Last Day of Classes	Final Exam
Sunday, 9 November, 2025	Sunday, 14 December 2025	Sunday, 21 December 2025

Score Distribution:

Class Test / Quizzes	Attendance	Performance	Assignment/ Presentation/Viva	Mid Exam	Final Exam
20%	5%	5%	10%	30%	30%

Grading Policy:

Marks (%)		LG	GP
80% or above	:	A+	4.00
75% to less than 80%	:	A	3.75
70% to less than 75%	:	A-	3.50
65% to less than 70%	:	B+	3.25
60% to less than 65%	:	B	3.00

Marks (%)		LG	GP
55% to less than 60%	:	B-	2.75
50% to less than 55%	:	C+	2.50
45% to less than 50%	:	C	2.25
40% to less than 45%	:	D	2.00
Less than 40%	:	F	0.00

Special Instructions:

- Students are not allowed to enter into the classroom after 5 minutes of starting time.
- No make-up quizzes and assignment will be held.
- Students are requested to switch off their mobile telephone during the class hour.
- There is zero tolerance for cheating at EWU. Students caught with cheat sheets in their possession, whether used or not used, copying from cheat sheets, writing on the palm of hand, back of calculators, chairs or nearby walls etc. would be treated as cheating in the exam hall. The only penalty for cheating is expulsion from EWU.



28-09-2025
(Prof. Dr. Md. Shariful Alam)